

PHASE 8 — Geo-Spatial & Location Intelligence

Real Estate AI Decision Intelligence Platform

Follows Phase 5 in the Master A-Z Prompt's 10-phase architecture (runs in parallel with Phases 6/7 — all three depend only on Phase 1, not on each other; they integrate in Phase 9). Matches the established code-structure convention: live PostgreSQL via `db/connection.py`, `src/<topic>/*.py` modules, three orchestrators, an automated test suite.

Scale note: this phase operates at the **locality level (90 rows)**, not the property level (8,000 rows) — so every result covers the full portfolio, no demo sampling needed like Phases 6/7 used.

What was built

Module	Purpose	Master Prompt Section
<code>data_loader.py</code>	Sources every OBSERVED input (locality scores, infrastructure records, market history) from Phase 1 tables.	8.2
<code>accessibility.py</code>	Distance-weighted accessibility, infrastructure, and neighborhood-quality sub-scores from real infrastructure distances.	8.4, 8.5, 8.6
<code>geo_risk.py</code>	flood/environmental risk correctly INSUFFICIENT DATA (no such data exists); infrastructure delivery-risk and concentration-risk as documented proxy indicators.	8.7
<code>location_score.py</code>	7-component weighted composite with documented weights and renormalization.	8.8
<code>geo_engine.py</code>	Top-level GeoResult orchestrator.	8.9
<code>batch_runner.py</code>	Runs the full 90-locality portfolio.	—
<code>export_results_csv.py</code>	4 numbered CSVs → <code>docs/geo_results_csv/</code> .	—

generate_charts.py	4 charts → docs/geo_charts/.	—
test_phase8.py	26 automated tests (all passing).	—

Full detail: [docs/phase8_report.md](#), [docs/limitations.md](#).

Notable finding

flood_risk and environmental_risk are INSUFFICIENT DATA for all 90 of 90 localities — this platform has no hazard/climate data anywhere, and the engine correctly refuses to estimate a substitute rather than inventing one (Master Prompt §8.7). A real normalization bug was also found and fixed during development: the first version of the market-strength formula saturated 78% of localities at a perfect score because this dataset's demand/supply ratios run structurally above the assumed ceiling. Fixed with data-driven P5/P95 normalization anchors; locked in with a regression test. Details in [docs/phase8_report.md](#) Sections 5–6.

How to run

With a live database (production path)

```
cd phase8
pip install -r requirements-phase8.txt --break-system-packages
export DB_USER=...
export DB_PASSWORD=...

python3 src/geo/export_results_csv.py      #
docs/geo_results_csv/*.csv
python3 src/geo/generate_charts.py        # docs/geo_charts/*.png
```

Local re-check (no database — what was used to produce this deliverable)

```
cd phase8
pip install -r requirements-phase8.txt --break-system-packages

python3 src/geo/export_results_csv.py --source csv
python3 src/geo/generate_charts.py --source csv

python3 -m pytest src/geo/test_phase8.py -q
```

Single-locality example

```
from src.geo.geo_engine import run_geo_engine

result = run_geo_engine(locality_id=2, source="csv")
print(result["location_score"]["location_score"])
print(result["geo_risk"]["flood_risk"])          # INSUFFICIENT
DATA -- honest, not invented
print(result["accessibility"]["accessibility_score"])
```